



The gastrointestinal tract: properties and role in allogeneic hematopoietic stem cell transplantation

Ilias Pessach, Panagiotis Tsirigotis & Arnon Nagler

To cite this article: Ilias Pessach, Panagiotis Tsirigotis & Arnon Nagler (2017) The gastrointestinal tract: properties and role in allogeneic hematopoietic stem cell transplantation, Expert Review of Hematology, 10:4, 315-326, DOI: [10.1080/17474086.2017.1288566](https://doi.org/10.1080/17474086.2017.1288566)

To link to this article: <https://doi.org/10.1080/17474086.2017.1288566>



Published online: 08 Feb 2017.



Submit your article to this journal [↗](#)



Article views: 538



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 2 View citing articles [↗](#)

REVIEW



The gastrointestinal tract: properties and role in allogeneic hematopoietic stem cell transplantation

Ilias Pessach^a, Panagiotis Tsigotis^a and Arnon Nagler^b

^aSecond Department of Internal Medicine, Division of Hematology, ATTIKO University Hospital, National and Kapodistrian University of Athens, Athens, Greece; ^bDivision of Hematology and Bone Marrow Transplantation, Chaim Sheba Medical Center, Tel Hashomer, Israel

ABSTRACT

Introduction: The GI-tract is a major target for both the intensive chemo and/or radiotherapy conditioning as well as for GVHD and therefore is closely associated with transplant outcome. Apart from being a target, the GI-tract is also a mediator and therefore is also a key player of the pathogenetic process following allogeneic transplantation.

Areas covered: The intestinal homeostasis is regulated through complicated interactions between the key players of this process which are the intestinal epithelium, the intestinal immune system, and the intestinal microbiota. A brief description of these elements, based on published english-language articles in PubMed, as well as their role during the process of allo-HSCT is discussed in this review.

Expert commentary: Data on GI-tract properties suggest a central role for the intestine in regulation of immunity, both in healthy – steady state conditions and in pathological states such as during allo-HSCT. Given the fact that in the allogeneic transplant setting severe complications such as infections and GVHD are limiting this treatment modality, understanding the mechanisms that mediate intestinal homeostasis could lead to new preventive methods and improved outcomes.

ARTICLE HISTORY

Received 12 September 2016
Accepted 26 January 2017

KEYWORDS

GI-tract; allo-HSCT; GVHD; paneth cells; Mast Cells (MCs); Innate Lymphoid Cells (ILCs); Antigen Presenting Cells (APCs); Toll-like receptors (TLRs); NOD-like receptors (NLRs); intestinal microbiota

1. Introduction

In spite of the high curative potential and the progress made, allogeneic hematopoietic stem cell transplantation (allo-HSCT) is still limited by significant morbidity, mainly due to infections and graft-versus-host disease (GVHD) [1]. Gastrointestinal tract (GI-tract) is a major target for both the intensive chemo and/or radiotherapy conditioning as well as for GVHD and therefore is closely associated with transplant outcome. The disruption of the intestinal barrier during conditioning results in the translocation of bacteria and bacterial products from the intestinal lumen to the bloodstream. As a result, besides the development of sepsis, an allo-reactive process through interactions with receptors such as Toll-like receptors (TLRs) expressed by immune cells, is triggered. However, the GI-tract is not only a target but also a mediator and therefore is a key player of the pathogenetic process following allogeneic transplantation [2]. Therefore, further understanding of the GI-tract properties and their role during allo-HSCT may lead to the development of novel methods for maintenance of intestinal epithelium integrity and reduced incidence of infections and GVHD.

The GI-tract is covered by the intestinal epithelium which is the largest surface area present in human body and serves as a barrier protecting the organism from the trillions of microbes present in the intestinal lumen while is also responsible for the absorption of nutrients daily provided by the food. For the accomplishment of these tasks, the intestinal epithelium is equipped by highly specialized cells and by a large amount of immune cells collectively known as the gut-

associated lymphoid tissue (GALT). A brief description of these elements as well as their role during the process of allo-HSCT is discussed in this review.

2. Intestinal epithelium

2.1. Properties and functions

The GI-tract is continuously in contact with a huge amount of antigens including diverse commensal microbiota, food antigens, and potentially pathogenic microbes, and thus requiring both specific adaptive and nonspecific fast innate responses in order to maintain intestinal homeostasis (Figure 1) [3]. Intestinal mucosa is not only simply a structural barrier but also an organ made up of specialized intestinal epithelial cells (IECs). IECs however are short lived, and constant replenishment of the intestinal epithelium is mediated by intestinal stem cells (ISCs) located at the base of intestinal crypts. ISCs give rise to all subsets of differentiated IECs including enterocytes, entero-endocrine cells, Paneth, and goblet cells. The enterocytes are responsible for the absorption of nutrient molecules into the body, while goblet cells produce mucins and are responsible for the generation of a mucus layer covering the intestinal epithelium and separating it from the contents of the intestinal lumen. Paneth cells are located within the crypts and produce antimicrobial peptides (AMPs) [4,5].

Intestinal mucosa is also equipped with GALT. GALT consists of an enormous amount of immune tissue including (1) intraepithelial lymphocytes (IELs), (2) lymphoid and myeloid

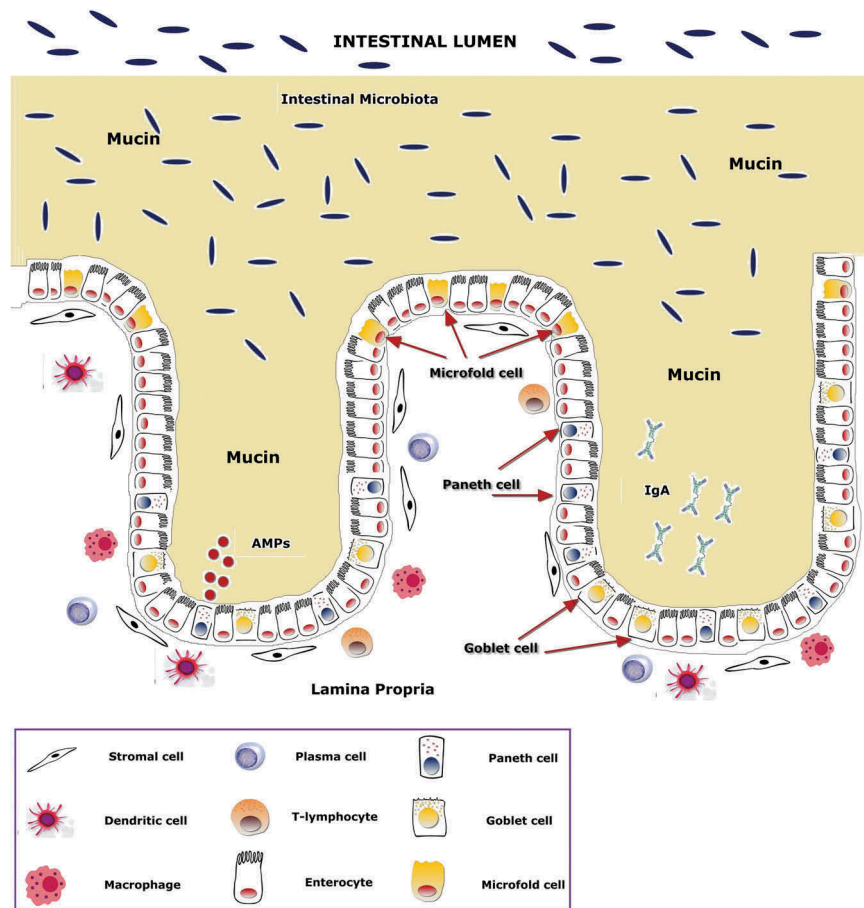


Figure 1. Human intestinal microbiota contains trillions of bacteria with the vast majority of them being harmless commensals living in a state of harmonic symbiosis with the host. However, intestinal epithelium is also exposed to large number of pathogens and intestinal homeostasis is mediated by certain lines of defence including the intestinal epithelial barrier and the intestinal immune system. Intestinal epithelial cells (IECs) are short-lived and constant replenishment of intestinal epithelium is mediated by intestinal stem cells (ISCs) that locate at the base of intestinal crypts. ISCs give rise to all subsets of differentiated IECs including enterocytes, Paneth, and goblet cells. Goblet cells produce mucins and are responsible for the generation of a mucus layer covering the intestinal epithelium and separating it from the contents of the intestinal lumen. Paneth cells are located within the crypts and produce antimicrobial peptides (AMPs). Microbes as well as microbial antigens crossing the intestinal barrier are phagocytosed by phagocytic cells located at Lamina Propria (LP) such as macrophages (MP) and dendritic cells (DCs). The decision of the immune system to react against pathogenic invaders or to ignore harmless commensals is mediated by different subsets of MPs and DCs. Antigen-specific T-cell with a memory phenotype as well as IgA secreting plasma cells are residents of LP. IgA is transported across the epithelial barrier into the mucous layer of the intestinal epithelial to serve as another line of defence against pathogens.

cells located in the lamina propria (LP), and (3) organized secondary lymphoid structures, such as isolated lymphoid follicles, Peyer's patches (PPs), and mesenteric lymph nodes (MLNs). Activated immune effectors, such as T cells with cytotoxic or helper activity and plasma cells producing antibodies, are mainly located in the intestinal epithelium and the LP, whereas PPs and MLNs are the sites for induction of primary immune responses to foreign antigens [6].

The intestinal immune system has a dual task: to elicit protective immune responses against pathogens that potentially may cause harm to the host, while at the same time to promote and maintain tolerance to harmless commensals and food proteins and therefore protect the host from inappropriate and useless inflammatory responses. Antigen capture, processing and presentation to immune cells, is mediated by antigen-presenting cells (APCs), such as macrophages and dendritic cells (DCs) [3].

GALT is covered by a specialized form of follicle-associated epithelia (FAE) consisting of various cell types including microfold (M) cells. M cells are highly specialized IECs with

phagocytic activity and their main function is transfer of macromolecules, antigens, and microbes from the gut lumen to intestinal submucosa through phagocytosis and transcytosis. Following transfer through FAE, antigenic molecules are taken up by DCs for processing and presentation to cells of the adaptive immune system [5].

According to many studies, M cells are sampling several microbes and antigens through 'immunosurveillance' receptors expressed on their apical surfaces. Such M-cell-specific proteins are glycoprotein 2, uromodulin, cellular prion protein, ANXA5, and peptidoglycan recognition protein-1 [7–10]. Furthermore, studies also show that in the absence of M cells or antigen sampling by them, antigen-specific T-cell responses in the PPs of orally infected mice are absent or severely diminished [7,11]. Thus, sampling of gut luminal antigens through M cells is important for induction of mucosal immune responses [7,10–12].

The function as well as the role of specific cell subsets in gut homeostasis during allogeneic transplantation is discussed below.

2.2. Paneth cells

Paneth cells are located in the intestinal crypts of the small intestine in close adjustment to intestinal epithelial stem cells. Microscopically, they are characterized by the presence of cytoplasmic eosinophilic granules containing AMPs. The principal AMPs secreted by Paneth cells are alpha-defensins, which are known as cryptdins in mice [13]. Defensins have a potent antimicrobial activity while being nontoxic to surrounding host epithelial cells [14]. Paneth cells are stimulated to secrete defensins after exposure to bacteria or to bacterial products such as lipopolysaccharide (LPS), thereby contributing to maintenance of the GI homeostasis. Importantly, AMPs secreted by Paneth cells have selective bactericidal activity against pathogens such as *Klebsiella*, *Salmonella*, and *Escherichia coli*, while bactericidal activity against commensals is minimal to absent [15,16].

Paneth cells are targeted by the GVHD process and recent animal and human data have shown that these cells are markedly reduced in recipients suffering from intestinal GVHD. Indeed in an animal model of allogeneic SCT, analysis of data showed that fecal levels of defensins were markedly reduced in mice with GVHD as compared to control mice (mice transplanted with T-cell depleted marrow). Moreover, intestinal microbiota diversity was also significantly reduced in GVHD mice with *E. coli* dominance while intestinal microbiota was unaffected in control mice [17].

Paneth cells can be identified and quantified in biopsy samples by using light microscopy and lysozyme staining. Recently, Paneth cell quantification was performed in duodenal biopsy samples of 116 patients with intestinal acute GVHD. The number of Paneth cells per high power field (HPF) at diagnosis of GVHD was inversely associated with the clinical severity of GVHD. In multivariate analysis, patients with a low number of Paneth cells (<4 per HPF) had a significantly increased non relapse mortality (NRM) in comparison with patients with high numbers of duodenal Paneth cells (>4 per HPF). Authors concluded that the enumeration of duodenal Paneth cells is a useful and readily reproducible marker of disease severity that can be used for estimation of outcome after GVHD development [18].

In line with previous data is the finding of regenerating islet-derived 3-alpha (Reg3a) as a biomarker of GI GVHD. Reg3a is a C-type lectin with antimicrobial activity produced by Paneth cells. Reg3a is secreted in the intestinal lumen and besides its antimicrobial activity, Reg3a also acts on intestinal epithelium by inducing an antiapoptotic effect on IECs and thus helps maintaining intestinal integrity. Elevated plasma levels of Reg3a are characteristic of intestinal GVHD and help in the differential diagnosis from infection associated diarrhea. Importantly, there is a direct association of plasma levels of Reg3a with the clinical severity of intestinal GVHD. The correlation of mucosal damage with plasma REG3 levels suggests that disruption of the mucosal epithelial barrier caused by GVHD allows the translocation of REG3 from the intestinal lumen to the bloodstream [19].

2.3. Mast cells

Mast cells (MCs) in the GI-tract regulate vascular and epithelial permeability, ion secretion, angiogenesis, peristalsis, fibrosis

and tissue repair, innate and adaptive immunity, bacterial defense, and chemotaxis [20,21]. Hence, MCs have an immunomodulatory role and uncontrolled or dysregulation of MC activation may hinder gut homeostasis and promote inflammation [20,21].

The fact that MCs can have an immunomodulatory function was further supported in a recent study investigating their role in HSCT [22]. In that study, C57BL/6-Kit^{W-sh/W-sh} recipient mice lacking MCs underwent allo-HSCT and their outcome was compared to wild-type mice. GVHD was severely exacerbated in C57BL/6-Kit^{W-sh/W-sh} mice and so was the mortality risk with a median survival time in C57BL/6-Kit^{W-sh/W-sh} mice of 13 versus 60 days in wild-type mice ($P < .0001$). The immunomodulating effect of MCs was further supported by the observation that the survival of C57BL/6-Kit^{W-sh/W-sh} mice during GVHD was significantly improved when they were engrafted with bone marrow-derived cultured MCs from C57BL/6 wild-type mice [22].

2.4. Innate lymphoid cells

Innate lymphoid cells (ILCs) belong to a lymphocytic subtype which in contrast to T and B cells lack antigen-specific receptors and B or T lineage-specific markers. ILCs are immune effectors with a major contribution to the first line of defense against pathogens [23]. Furthermore, ILCs have a significant immunomodulating and tissue regenerating role, especially at the mucosal surfaces by strengthening the epithelial cell junctions [24,25].

Cytotoxic natural killer (NK) cells are one of the members of the ILC family, discovered during the 1970s. In addition to NK, three other major subsets of ILCs have been discovered during the last decades, and their classification is based on the different functions mediated through a specific cytokine secretion profile [23].

ILC type 1 (ILC1) secrete IFN γ and TNF α in response to stimulation with IL-12. Human ILC1 subset is further divided in two populations based on IL-7 receptor A (IL-7Ra, CD127). The CD127-negative ILC1 subpopulation is an intraepithelial resident in human intestine with strong cytotoxic activity against intracellular pathogens such as *Salmonella typhimurium*. Instead, the CD127-positive ILC1 subpopulation, which is not cytotoxic, is mainly found in the gut LP and secretes IFN γ and TNF in response to intestinal infections from various pathogens such as *Toxoplasma gondii* [26,27].

ILC2 produce a variety of pro-inflammatory cytokines. More importantly, they secrete amphiregulin, which is a substance necessary for tissue regeneration. ILC2 are found among peripheral blood circulating ILCs and are also located in areas of chronic inflammation such as in nasal polyps of patients with chronic rhinosinusitis [28,29].

ILC3 are characterized by the production of IL-22 and IL-17 in response to various stimuli [30,31]. Based on the expression of the natural cytotoxicity receptor NKp44, ILC3 are classified in two different subgroups. NKp44-negative ILC3 are the most abundant ILC subgroup in human peripheral blood and secrete the pro-inflammatory cytokine IL-17. On the contrary, NKp44-positive ILC3 are producers of the tissue protective cytokine IL-22. Large numbers of these cells are residents of

the healthy human intestine and have an essential role in maintaining intestinal homeostasis [32]. Moreover, the IL-22 triggers the release of AMPs and promotes epithelial cell proliferation, thereby preserving the integrity of the epithelial barrier [33].

The significance of ILCs for GVHD protection has been shown in an animal model. IL-22 deficient mice suffered from severe intestinal GVHD after allogeneic transplantation. Histology showed excessive epithelial cell apoptosis resulting in intestinal barrier disruption [34]. Human studies also demonstrated the protective activity of ILC3 against GVHD. Patients with increased circulating numbers of NKp44+ ILC3, 12 weeks after transplantation, had low incidence of intestinal GVHD. Moreover, increased expression of gut homing markers such as CD69 on ILC3 was found to be associated with decreased severity and incidence of GVHD [35].

3. Methods for improving the intestinal epithelium integrity

As already mentioned, the intestinal mucosa is severely damaged due to both the toxicity of the conditioning regimen as well as the intestinal GVHD. Mucositis besides of being an extremely unpleasant situation for the patient is also the cause for other serious events such as bacterial translocation and sepsis. Factors with a known protective or even a regenerative effect on intestinal epithelium are the keratinocyte growth factor (KGF), IL-22, and R-spondin are thus of clinical importance.

The KGF, administered as part of a randomized trial, showed a remarkable efficacy in reducing oral mucositis in patients that underwent autologous SCT following a TBI-based conditioning [36]. Another prospective trial assessed the effect of KGF treatment on intestinal mucositis again in patients receiving auto-SCT. The results from KGF administration were compared with a similar group of consecutive patients who underwent auto-SCT in the same transplant center before the marketing authorization of KGF. Intestinal mucositis was evaluated by using standard scoring systems based on the presence and severity of clinical symptoms including pain, diarrhea, and others. In this trial, analysis of data clearly showed that the severity of intestinal mucositis was significantly decreased in the group of patients treated with KGF. Moreover, by using video-capsule endoscopy, it was observed that KGF-treated patients had macroscopically normal appearing small intestinal mucosa, while significant macroscopic lesions such as erythema, edema, erosions, and ulcers were observed in non-treated patients. Remarkably, a higher incidence of bacteremias was also observed in non-treated patients as compared to those treated with KGF [37].

The intestinal protective efficacy of KGF has been also tested in the setting of allo-HSCT with the aim to prevent intestinal and systemic GVHD. However, in a prospective randomized trial, KGF administration failed to show a GVHD-preventing effect [38].

The regenerating effect of exogenous administration of IL-22 on intestinal epithelium has also been shown in an animal model of GVHD. Mice treated with IL-22 after allogeneic transplantation had decreased incidence of intestinal GVHD and

reduced mortality as compared to mice treated with placebo. IL-22-treated mice showed enhanced recovery of ISC and reduced intestinal pathology scores [39].

4. APCs

Immune system reacts to invasion by a pathogen with an 'inflammatory response' intended to eliminate the invader. The activation of the immune system is initiated by signals delivered through pathogen recognition receptors (PRRs) after interaction with their ligands [40]. Ligands binding to PRRs are evolutionarily conserved molecules expressed by microbes, parasites, fungi, viruses known as pathogen-associated molecular patterns (PAMPs) [41].

During the last decades, many different classes of PRRs have been discovered, including Toll-like receptors (TLRs), retinoic acid inducible gene I-like receptors, and nucleotide oligomerization domain (NOD)-like receptors (NLRs) [42].

Activation of innate immunity is rapidly followed by an adaptive immune response, mainly mediated by APCs [43]. The most efficient APCs in the body are the DCs which express a large number of PRRs. Following interaction with the respective PAMPs, the DCs undergo a functional transformation and acquire the capacity to interact with naive antigen-specific immune cells [43].

DCs activated by PAMPs present pathogen-derived peptides in the context of major histocompatibility complex (MHC) molecules [44]. The complex MHC-peptide binds to T-cell receptors expressed on naive pathogen-specific T cells and this interaction delivers the first activating signal to the T cell and is called 'signal 1.' Finally, activated DCs also express a variety of costimulatory molecules, which bind to cognate receptors expressed on the surface of T cells and transmit more activating signals (referred as signal 2) required for the full activation of naive pathogen-specific T cells [44]. Therefore, activated CX3CR1+ DCs are necessary for an effective immune response and DC activation is induced after exposure to the so-called danger signals provided by the interaction between PRRs and PAMPs [45]. On the contrary, resting CD103+ DCs (not activated by exposure to PAMPs) have a tolerogenic potential promoting T-cell anergy after engagement with an antigen-specific naive immune cell. Peripheral tolerance to autoantigens is mediated through resting DCs since DCs not exposed to PAMPs are usually loaded with self-antigens [46].

5. Pattern recognition receptors

5.1. TLRs

Toll-like receptors (TLRs) are membrane receptors expressed in APCs that recognize pathogen-derived molecules after disruption of skin and/or mucosal surfaces [47]. Ligands for TLRs are bacterial lipoprotein for TLR1, lipoteichoic acid (gram-positive bacteria derived) for TLR2, double-stranded viral RNA for TLR3, LPS (gram-negative bacteria derived) for TLR4, bacterial flagellin for TLR5, single-stranded viral RNA for TLR7, phagocytized bacterial RNA for TLR8, unmethylated CpG islands (bacterial and viral DNA derived) for TLR9 [48]. TLR downstream

signaling activates a complex intracellular cascade, which eventually leads to an immune response of the host against the violating pathogen [47].

Convincing evidence on how important PAMP–TLR interactions are in the pathogenesis of GVHD derives from experimental rat models of GVHD. LPS – a structural component of gram-negative bacteria – is a potent stimulator of TLR4. After allo-HSCT, translocation of LPS and microorganisms from the intestinal lumen to the systemic circulation occurs as a result of the intestinal barrier injury by the conditioning regimen. LPS stimulates the secretion of the so-called cytokine storm leading to increased GVHD mortality [49]. Cooke and colleagues studied the therapeutic efficacy of LPS antagonism on GVHD incidence and severity by administration of a lipid-A analog after allogeneic bone marrow transplantation (allo-BMT). They found that LPS antagonism resulted in reduced intestinal and systemic GVHD [50]. In line with these findings are the observations that post allo-BMT-elevated LPS serum levels, positively correlated with histopathologic target organ damage in recipients with GVHD [51,52].

However and despite the importance of the LPS-TLR4 pathway in GVHD pathogenesis, animal models using TLR4-deficient mice showed that TLR signaling is not an absolute prerequisite for GVHD development [53,54]. This might be attributed to the existence of alternative (non-TLR) activation pathways of APCs. Similar observations have been reproduced in human clinical trials. Lorenz et al. tested the impact of 2 TLR4 polymorphisms (Asp299Gly and Thr399Ile) on the outcome of 237 patients after allo-HSCT from matched sibling donors [55]. The authors found that Asp299Gly and Thr399Ile polymorphisms are associated with a reduced signaling activity after TLR4 stimulation. However, the presence of polymorphisms in recipients or donors was not associated with significant differences in the incidence and severity of GVHD [55]. In a similar study, Elmaagacli et al. also found no association of the presence of TLR4 polymorphisms with the incidence of GVHD, in a group of 307 allo-HSCT patients transplanted from matched related or unrelated donor [56]. The same authors examined the effect of the T1486C and T1237C polymorphisms of the TLR9 gene on the outcome of 413 patients after allo-HSCT. T1486C and T1237C polymorphisms were reported to be associated with reduced receptor expression and were notably not associated with the incidence and/or severity of GVHD [57].

Lastly, the role of TLRs in GvHD is further complicated by the fact that TLR mediates not only stimulating but also tolerogenic signals to immune cells [58]. Consistent with this observation are data from animal models showing the immunomodulating effect of stimulating certain TLRs [59,60]. Indeed, the stimulation of TLR2 or TLR3, instead of augmenting the inflammatory process, protects mice from experimental colitis [61]. Similar data from a mouse model of experimental allergic encephalomyelitis have shown that stimulation of TLR7 with a synthetic agonist induced tolerance and reduced inflammation [62]. Furthermore, exogenous administration of the TLR5 ligand flagellin resulted in reduced severity of GVHD [59]. Similar effects were produced after administration of a TLR7 agonist [60].

5.2. NLRs

The nucleotide-binding oligomerization domain-like receptors (NLRs) are intracellular receptors binding to PAMPs that enter the cell via phagocytosis. NLRs cooperate with TLRs and regulate inflammation in response to pathogen invasion or cellular damage [63]. NLRs are expressed on APCs such as macrophages and DCs as well as on nonimmune cells such as epithelial cells. NLR family contains several receptors with NOD1 and NOD2 being the most well studied. NOD1 and NOD2 sense bacterial peptidoglycan-derived molecules such as muramyl dipeptide and induce pro-inflammatory responses [63].

Paradoxically, stimulation of NOD2 had a negative impact on GVHD development and outcome in a mouse allogeneic transplantation model. Briefly, NOD2 deficiency in recipients was associated with increased incidence of GVHD [64]. These results are in accordance with previous studies showing increased susceptibility to experimental colitis in NOD2 deficient mice [65,66].

Similar findings have been observed in human clinical trials, examining the association between NOD2 single nucleotide polymorphisms (SNPs) and increased incidence and severity of GVHD. In these trials, certain NOD2 polymorphisms (SNPs 8 [Arg702Trp], 12 [Gly908Arg], and 13 [Leu1007fsinsC]) were associated with reduced downstream signaling activation in response to bacterial cell products. The prevalence of these SNPs was found to be increased among patients with Crohn's disease. Holler et al. showed that the presence of one of these SNPs in recipients or donors of allo-HSCT is associated with significantly increased incidence of GVHD [67,68].

6. Intestinal microbiota and allogeneic transplantation

6.1. Intestinal microbiota and intestinal homeostasis

The term intestinal microbiota is used to define the ecological community of commensal and pathogenic microorganisms that colonize the human gut from mouth to anus. Microbiota consists from trillions of microbes, a number well above the number of human cells [69]. Since most of these microorganisms are not cultured by using standard conventional techniques, studying and identification of human microbiota depends on molecular methods based on the analysis of the small subunit ribosomal RNA (SSU rRNA) gene sequences which serve as markers for microbial taxonomy. Thus, bacteria are sub-classified in phyla, classes, orders, families, genus, and species.

However, only a few phyla are represented in human microbiota, with *Firmicutes* (consisting mainly of gram-positive clostridia), *Bacteroidetes* (consisting mainly of gram-negative bacteria), *Actinobacteria*, and *Proteobacteria* being the most prevalent [70,71]. Regardless of taxonomic differences, the microbiota that colonizes the human digestive track is innocent residents and under normal circumstances do not cause disease. Moreover, there is a dynamic interplay between these commensals and human cells which result in a mutual usefulness [69–71]. Currently, many of these functions of the human

normal microbiota are not known and this is an area of active research. The composition of microbiota is specific for each individual and depends on various parameters mainly acting throughout the first years of life [72]. After development, microbiota remains stable for large periods of time unless disturbed by exogenous factors such as infectious diseases, administration of antibiotics, inflammation, nutritional status, etc. [72].

Under normal conditions, microbiota influences the development of immune system during the first years of life and vice versa. As a proof of principle of this mutual influence, animal data have shown that mice growing under germ-free conditions do not develop normal immune system and are susceptible to microbial infections [73]. Moreover, other data from animal studies have shown that the presence of intestinal TH17 cells as well as of regulatory T cells (TRegs) are greatly reduced in antibiotic-treated or germ-free mice, showing that intestinal microbiota has a crucial role in the development of specific immune cell subsets [74–79]. Indeed, segmented filamentous bacteria promote the generation of TH17 cells in mice [74–76], while specific populations of commensal bacteria, such as 46 *Clostridium* spp. cluster IV and XIVa strains, and the human commensal bacterium *Bacteroides fragilis*, induce Tregs [77–79]. The mechanism by which specific bacterial populations induce the development of specific immune cell subsets remains poorly understood.

In addition to the above, intestinal microbiota not only affects the development of intestinal immune system, but also microbiota composition is affected by immune system. As an example, IgA deficiency has a major impact on microbiota development and composition [80,81], while intestinal microbiota is also a major regulator of IgA production, as the number of IgA producing cells in the intestine is markedly decreased in germ-free mice [82].

6.2. Intestinal microbiota and resistance to bacterial translocation

Germ-free mice are more susceptible to infection than conventionally raised mice [73], while antibiotic intake promotes pathogen colonization [83,84]. Therefore, microbiota could be protecting the host from infections.

A possible mechanism could be commensal bacteria competing with metabolically related pathogens for nutrients within the intestine, thus inhibiting their colonization. For example, *Escherichia coli* competes with enterohemorrhagic *E. coli* (an enteric pathogen known to cause substantial morbidity and mortality) for organic acids, amino acids, and other nutrients [85–87]. Another possible mechanism that has been proposed is the promoting effect of commensal bacteria to the mucin production. The epithelium under normal conditions produces mucins, forming a mucus layer as a physical barrier in order to inhibit pathogen invasion [88]. This mucus layer in germ-free mice has been found to be much thinner than in conventionally raised mice and should bacterial products such as LPS and peptidoglycan be administered, the mucus layer thickness is restored [89].

The microbiota of healthy persons shows a high diversity. Allo-HSCT, instead, has a major impact on the composition of

microbiota and usually results in loss of diversity and dominance of certain bacteria. A large prospective trial including 94 patients that underwent allo-HSCT was performed with the aim to identify parameters associated with the loss of microbial diversity and the dominance by certain microorganisms of the intestinal microbiota after transplantation [90]. Fecal samples were collected and studied before and after allo-HSCT. Dominance by a specific microorganism was defined the situation where a particular microbe constituted >30% of the total intestinal microbiota composition. Authors observed that the risk of intestinal domination by *Proteobacteria*, *Enterococcus*, and *Streptococcus* increased over time and a major contributing factor was administration of antimicrobial chemotherapy. As was anticipated, dominance by *Enterococcus* was strongly associated with treatment with metronidazole due to its inhibitory effect on anaerobes. Risk of bacteremia due to specific organisms was directly associated with dominance by certain species. As an example, domination by *Proteobacteria* increased the risk of developing gram-negative bacteremia. Regarding intestinal microbiota diversity, authors classified patients into low-, intermediate-, and high-diversity groups depending on the microbiota composition at the time of stem cell engraftment. Microbiota diversity was inversely associated with administration of antibiotics with specificity against anaerobic bacteria. Of note, in multivariate analysis, the degree of bacterial diversity was inversely correlated with NRM. The highest NRM (52%) was observed in patients within the low-diversity group, while NRM in the group of high-diversity patients was as low as 8%. Finally, patients with intermediate bacterial diversity had an intermediate NRM of 24% [90].

6.3. Intestinal microbiota and GVHD pathogenesis

Initial studies performed, few decades ago, in children and adults suffering from aplastic anemia showed that elimination of the intestinal bacteria has a positive impact on the outcome of allo-HSCT by reducing infections and severe acute GVHD (aGVHD) [91,92]. These clinical data were in accordance with animal models showing reduced incidence of GVHD and GVHD-related mortality in mice growing in germ-free conditions [93,94]. However, this strategy has not been adopted by many transplant centers because of the variability in the success rate of gut decontamination, as well as the high cost, and the technical difficulties [95]. Currently, the modulation of intestinal microbiota as a method for GVHD prevention is an issue of active and ongoing research.

The discovery of new techniques enabling accurate study of the intestinal microbial content resulted in the renewal of the interest in the topic of the effect of microbiota composition on transplant outcome. The association between GVHD and microbiota composition was firstly suggested in animal studies and was later showed in human trials as well. Recently, several studies have demonstrated a close association between intestinal microbiota composition – diversity and aGVHD.

In animal experiments, analysis of the microbiota composition indicated a dramatic loss of bacterial diversity in recipient

mice with aGVHD compared to recipient mice without aGVHD. GVHD was associated with an expansion of *Lactobacillus*, *Enterococcus*, *Escherichia*, and *Klebsiella*, accompanied by loss of anaerobes including members of the order *Clostridiales* [96].

Similarly, a human study has shown that the incidence of GVHD is associated with loss of microbiota diversity and particularly with an increase in *Lactobacillales* and a relative decrease in *Clostridiales* [96]. In another human study, researchers observed increased incidence and GVHD mortality in patients with dominance of microbiota by *Blautia* which belongs to the *Clostridia* class [97]. Taken together, previous studies have convincingly shown that GVHD has been associated with significant changes in intestinal microbiota composition and with loss of bacterial diversity [90]. However, these data should be interpreted cautiously and we should keep in mind that statistical association is not synonymous to causality.

Members of the intestinal microbiota metabolize substances of host origin as well as those contained in food. A group of by-products of this process is various short-chain fatty acids (SCFAs) including acetate, propionate, and butyrate. Intestinal epithelium and the organism as a whole are under the influence of the so-called Intestinal Metabolome consisting of many products derived from host metabolism, bacterial metabolism, and host-bacterial co-metabolism. The effect of intestinal metabolome on human organism is largely unknown, but it has been hypothesized that it contributes to the pathogenesis of various disorders. As an example, previous studies have shown that butyrate, a by-product of saccharide metabolism by *Clostridia*, has a tolerogenic effect on intestinal immune system mediated by the generation of intestinal Tregs. This effect has been attributed to the fact that butyrate is an inhibitor of histone deacetylase and, when present in high amounts in intestinal epithelium, induces epigenetic modification that is probably associated with the tolerogenic potential [98,99].

In an animal model of allogeneic MHC-mismatched transplantation, researchers were able to show that butyrate levels were significantly reduced in the intestinal tissue after allo-HSCT but not after syngeneic transplantation. Interestingly, it was shown that exogenous administration of high-dose butyrate resulted in increased butyrate levels, reduced GVHD pathology and mortality in allo-HSCT recipient mice. The protective effect of butyrate was attributed to reduced inflammation and reduced apoptosis of ISCs [100].

Since butyrate is a by-product of microbial metabolism, an attractive hypothesis is that high-butyrate producing microbiota might have a protective effect from GVHD mortality. Next, the effect of colonization with *Clostridia* on the outcome of allo-HSCT was examined. Mice were fed via intragastric gavage with selected *Clostridia* strains, with a known high-butyrate producing ability. Mice colonized with *Clostridia* had significantly increased intestinal butyrate levels, reduced GVHD pathology and mortality as compared to control mice after allo-HSCT. As a further proof of the

concept, prevention of colonization by *Clostridia*, by antibiotics active against anaerobes, resulted in increased GVHD severity and mortality [101].

7. Expert commentary

Understanding the mechanisms that mediate intestinal homeostasis during stress as is the case after allo-HSCT is mandatory for the development of novel methods for prevention of severe complications such as infections and GVHD. Intestinal homeostasis is regulated through complicated interactions between the key players of this process which are the intestinal epithelium, the intestinal immune system, and the intestinal microbiota. This topic has been the subject of intensive research and the current status of our knowledge is briefly mentioned below:

Paneth cells are highly specialized IECs secreting AMPs into the intestinal lumen. The principal AMPs secreted are defensins and the C-type lectin Reg3a [13,19]. Importantly, AMPs secreted by Paneth cells have selective bactericidal activity against pathogens, while being nontoxic against commensals or to surrounding host epithelial cells, and therefore are important regulators of the intestinal microbiota composition [15,16]. Paneth cells are targeted by the GVHD process and recent animal and human data showed that these cells are markedly reduced in recipients suffering from intestinal GVHD [17,18]. In addition, clinical trials in humans have shown that the degree of Paneth cell loss is directly associated with the severity and outcome of intestinal GVHD [18]. As for Reg3a, it can serve as a biomarker for the diagnosis and prognosis of intestinal GVHD: Paneth cell apoptosis and disruption of the intestinal barrier result in the translocation of Reg3a from the intestinal lumen to the bloodstream and therefore plasma levels of Reg3a are directly associated with the degree of Paneth cell loss and the damage to intestinal epithelium. Indeed, elevated plasma levels of Reg3a are helpful in differential diagnosis of intestinal GVHD from infectious diarrhea [19].

Intestinal MCs regulate vascular and epithelial permeability, ion secretion, angiogenesis, peristalsis, fibrosis and tissue repair, innate and adaptive immunity, bacterial defense, and chemotaxis [21]. Moreover, MCs have an important immunomodulatory role and animal studies have shown that mice deficient of MCs develop severe GVHD after allogeneic transplantation in contrast to wild-type mice [22]. The protective effect of MCs in GVHD has not been translated yet in to the clinic.

ILCs are immune effectors with a major contribution to the first line of defense against pathogens [23]. Furthermore, ILCs have a significant immunomodulatory and tissue regenerating role, especially at mucosal surfaces by strengthening the epithelial cell junctions [24,25]. A very interesting ILC3 subset that deserves further exploration is the NKp44-positive ILC3 which produce the tissue protective cytokine IL-22 and have an essential role in maintaining intestinal homeostasis [32]. IL-22 is an ISC growth factor, promotes epithelial cell proliferation, and

therefore is a major contributor for the maintenance of the epithelial barrier integrity [33]. In line with previous data is the observation that IL-22 deficient mice suffered from severe intestinal GVHD after allogeneic transplantation, as compared to wild-type mice. Mice treated with IL-22 after allogeneic transplantation had decreased incidence of intestinal GVHD and reduced mortality as compared to mice treated with placebo [34].

Intestinal immune system has a dual function; immune activation against pathogens, and immune tolerance to harmless commensals and food proteins in order to prevent inappropriate tissue inflammation and damage. This dual function is partly mediated through interactions of PAMPs with the relevant PRRs such as TLRs and NLRs expressed in various cell types [41]. The association of certain SNPs of either TLRs or NLRs with transplant outcome has been the subject of intensive research. Most of the studies examining the impact of TLR polymorphisms in the incidence and severity of GVHD after allo-HSCT failed to show any correlation [55–57]. On the contrary, other clinical trials examining NOD2 polymorphisms in donors and recipients showed that there is a strong association between the presence of polymorphisms and increased GVHD incidence and severity [67,68].

The term intestinal microbiota is used to define the ecological community of commensal and pathogenic microorganisms that colonize the human gut from mouth to anus [69]. The introduction of new molecular technologies based on the analysis of SSU rRNA gene sequences which serve as markers for microbial taxonomy paved the way for the detailed analysis of microbiota composition which was not possible by using the standard culture techniques. There is a dynamic interplay between microbiota and intestinal epithelium which under normal circumstances results in a mutual usefulness [69–71]. Currently, many of these functions of human normal microbiota are not known and this is an area of active research. The composition of microbiota is specific for each individual, depends on various parameters, and remains stable for large periods of time unless disturbed by exogenous factors such as infectious diseases, administration of antibiotics, inflammation, nutritional status, and others [72]. Normal intestinal microbiota is an effective barrier against colonization by pathogenic and resistant microbes. Normal microbiota contains an extremely large number of different microbes. Human clinical trials have shown that the high microbial diversity is associated with less infection and reduced NRM after allo-HSCT [90]. In addition, data from animal and human studies have demonstrated that GVHD affects the composition of microbiota and the dominance of certain bacteria has been associated with the incidence of GVHD [90]. The incidence of GVHD was shown to be associated with loss of microbiota diversity and particularly with an increase in *Lactobacillales* and a relative decrease in *Clostridiales* [96]. In contrast, dominance of microbiota by *Blautia* was shown to be associated with increased incidence and GVHD mortality [97].

8. Five-year view

Taken into consideration the fact that microbiota composition has a major effect on GVHD development, it is reasonable to assume that microbiota modulation might be proved an effective method for GVHD prevention.

As already mentioned, the effect of gut decontamination on GVHD development has been tested in past trials [102,103]. However, GI decontamination was not widely adopted mainly due to high costs, unsuccessful decontamination rates, and inconsistent results [104]. The rationale against the 'gut decontamination' approach is that this method induces an unselective decontamination without distinction between useful commensals and danger resistant pathogens [105]. Results, from previous trials on microbiota composition showing that loss of diversity results in increase rates of infection and GVHD mortality, set up the rationale for microbiota modulation by prebiotic, probiotic, and antibiotic as a method of GVHD prevention instead of unselective decontamination [106].

Microbiota modulation by the use of probiotics seems an attractive method. Indeed, animal models have shown that probiotic administration could be effective in reducing GVHD severity and mortality [107]. The major obstacle for the clinical use of probiotics in human trials is the safety issue and the potential harm when administered in immunosuppressed individuals. However, administration of *Lactobacillus plantarum* during conditioning and throughout the neutropenic period has been tested in 30 children and adolescents. Results, although preliminary, showed that administration of probiotics was safe. No bacteremia or infection death attributable to *Lactobacillus* was observed in any patient. Moreover, a probable beneficial effect on GVHD occurrence was suggested in this trial, which resulted in the interest for conducting larger trials to evaluate the safety and efficacy of probiotics [108].

Regarding prebiotics, these are food components usually not digested and with the ability to modulate intestinal microbiota [109]. As an example, oligosaccharides are indigestive components of human milk and act as soluble factors preventing the adhesion of enteropathogenic bacteria to intestinal epithelium [110]. The anti-infective and anti-inflammatory effect of prebiotics on intestinal epithelium deserves testing in future clinical trials with the aim to reduce GVHD.

Another interesting method for microbiota modulation is by fecal microbiota transplantation (FMT), performed by the infusion of feces from a healthy donor into the intestinal track of a patient. FMT has proven to be an effective method for treatment of *Clostridium difficile* infection, when all other methods have failed [111]. Similarly, eradication of colonization from multidrug resistant bacteria has been performed in a myeloma patient by using the method of FMT [112]. Theoretically, if a protective, against GVHD or other major complications, microbiota will be defined, then FMT could be a promising approach for altering microbiota toward a pre-defined composition. Although this area is still in an

embryonic phase of development, the idea of testing FMT as a method for GVHD prevention is tempting.

9. Key issues

- Intestinal homeostasis is regulated through complicated interactions between the intestinal epithelium, the intestinal immune system, and the intestinal microbiota – all key players of this process.
- Paneth cells are highly specialized IECs secreting defensins and the C-type lectin Reg3a. They are important regulators of the intestinal microbiota composition. Clinical trials in humans have shown that the degree of Paneth cell loss is directly associated with the severity and outcome of intestinal GVHD. Reg3a can serve as a biomarker for the diagnosis and prognosis of intestinal GVHD.
- Intestinal mast cells have an immunomodulatory function. Animal studies have shown that mice deficient of mast cells develop severe GVHD after allogeneic transplantation in contrast to wild type mice.
- ILC3 NKp44-positive cells located in LP, produce IL-22. IL-22 is an ISC growth factor. Exogenous administration of IL-22 reduced GVHD incidence and severity in a mouse model.
- Previous studies in humans have demonstrated that NOD2 polymorphisms in both donors and recipients were associated with increased GVHD incidence and severity.
- The introduction of new molecular technologies based on the analysis of SSU rRNA gene sequences paved the way for the detailed analysis of microbiota composition. The composition of microbiota is specific for each individual, and remains stable for large periods of time unless disturbed by exogenous factors.
- Intestinal microbiota with high microbial diversity is associated with less infection and reduced non-relapse mortality after allo-HSCT.
- Intestinal microbiota dominated by certain bacteria such as *Blautia* is associated with increased incidence and GVHD mortality.
- The effect of intestinal metabolome on human organism is largely unknown. It is postulated that contributes to the pathogenesis of various disorders. Previous studies have shown that butyrate which is a by-product of *Clostridia* metabolism, has a tolerogenic effect on intestinal immune system reducing the incidence of GVHD.

Acknowledgments

The initial idea and concept for the review was of A.N. Data and results were extracted by the first author (I.P.), and were crosschecked by P.T. and A.N. The first author (I.P.) was responsible for the main writing of the study, P.T. drafted the paper and revised it critically, and all authors revised the paper. P.T. drew the figure of the paper.

Funding

This paper was not funded.

Declaration of interest

The authors have no relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript. This includes employment, consultancies, honoraria, stock ownership or options, expert testimony, grants or patents received or pending, or royalties.

References

Papers of special note have been highlighted as either of interest (•) or of considerable interest (••) to readers.

1. Thomas ED, Storb R, Clift RA, et al. Bone marrow transplantation. *N Engl J Med.* 1975;292:832–843.
2. Copelan EA. Hematopoietic stem-cell transplantation. *N Engl J Med.* 2006;354:1813–1826.
3. Hooper LV, Littman DR, Macpherson AJ. Interactions between the microbiota and the immune system. *Science.* 2012;336:1268–1273.
4. Pelaseyed T, Bergstroöm JH, Gustafsson JK, et al. The mucus and mucins of the goblet cells and enterocytes provide the first defense line of the gastrointestinal tract and interact with the immune system. *Immunol Rev.* 2014;260:8–20.
5. Goto Y, Ivanov II. Intestinal epithelial cells as mediators of the commensal-host immune crosstalk. *Immunol Cell Biol.* 2013;9:204–214.
6. Pabst O, Mowat AM. Oral tolerance to food protein. *Mucosal Immunol.* 2012;5:232–239.
7. Hase K, Kawano K, Nochi T, et al. Uptake through glycoprotein 2 of FimH⁺ bacteria by M cells initiates mucosal immune responses. *Nature.* 2009;462:226–231.
8. Nakato G, Hase K, Suzuki M, et al. Cutting edge: brucella abortus exploits a cellular prion protein on intestinal M cells as an invasive receptor. *J. Immunol.* 2012;189:1540–1544.
9. Rand JH, Wu XX, Lin EY, et al. Annexin A5 binds to lipopolysaccharide and reduces its endotoxin activity. *Mbio.* 2012;3:e00292–00311.
10. Wang J, Gusti V, Saraswati A, et al. Convergent and divergent development among M cell lineages in mouse mucosal epithelium. *J. Immunol.* 2011;187:5277–5285.
11. Kanaya T, Hase K, Takahashi D, et al. The ets transcription factor spi-B is essential for the differentiation of intestinal microfold cells. *Nat. Immunol.* 2012;13:729–736.
12. Lelouard H, Fallet M, de Bovis B, et al. Peyer's patch dendritic cells sample antigens by extending dendrites through M cell-specific transcellular pores. *Gastroenterology.* 2012;142:592–601.
13. Ayabe T, Satchell DP, Wilson CL, et al. Secretion of microbicidal alpha-defensins by intestinal paneth cells in response to bacteria. *Nat Immunol.* 2000;1:113–118.
14. Eriguchi Y, Takashima S, Oka H, et al. Graft-versus-host disease disrupts intestinal microbial ecology by inhibiting paneth cell production of alpha-defensins. *Blood.* 2012;120:223–231.
15. Ouellette AJ, Hsieh MM, Nosek MT, et al. Mouse paneth cell defensins: primary structures and antibacterial activities of numerous cryptdin isoforms. *Infect Immun.* 1994;62:5040–5047.
16. Masuda K, Sakai N, Nakamura K, et al. Bactericidal activity of mouse alpha-defensin cryptdin-4 predominantly affects noncommensal bacteria. *J Innate Immun.* 2011;3:315–326.
17. Eriguchi Y, Nakamura K, Hashimoto D, et al. Decreased secretion of paneth cell α-defensins in graft-versus-host disease. *Transpl Infect Dis.* 2015;17:702–706.
18. Levine JE, Huber E, Hammer ST, et al. Low paneth cell numbers at onset of gastrointestinal graft-versus-host disease identify patients at high risk for non relapse mortality. *Blood.* 2013;122:1505–1509.
- **Recent study evaluating Paneth cell loss with intestinal GVHD severity and outcome.**
19. Ferrara JL, Harris AC, Greenson JK, et al. Regenerating islet-derived 3-α is a biomarker of gastrointestinal graft-versus-host disease. *Blood.* 2011;118:6702–6708.
- **Recent study showing that Reg3a can serve as a biomarker for the diagnosis and prognosis of intestinal GVHD.**

20. Reber LL, Sibillano R, Mukai K, et al. Potential effector and immunoregulatory functions of mast cells in mucosal immunity. *Mucosal Immunol.* 2015;8:444–463.
21. Bischoff SC. Role of mast cells in allergic and non-allergic immune responses: comparison of human and murine data. *Nat Rev Immunol.* 2007;7:93–104.
22. Leveson-Gower DB, Sega EI, Kalesnikoff J, et al. Mast cells suppress murine GVHD in a mechanism independent of CD4+CD25+ regulatory T cells. *Blood.* 2013;122:3659–3665.
- **Recent study assessing GVHD in mice lacking mast cells compared with wild-type mice.**
23. Spits H, Artis D, Colonna M, et al. Innate lymphoid cells – a proposal for uniform nomenclature. *Nat Rev Immunol.* 2013;13:145–149.
24. Satoh-Takayama N, Vosschenrich CA, Lesjean-Pottier S, et al. Microbial flora drives interleukin 22 production in intestinal NKp46⁺ cells that provide innate mucosal immune defense. *Immunity.* 2008;29:958–970.
25. Sonnenberg GF, Fouser LA, Artis D. Border patrol: regulation of immunity, inflammation and tissue homeostasis at barrier surfaces by IL-22. *Nat Immunol.* 2011;12:383–390.
26. Fuchs A, Vermi W, Lee JS, et al. Intraepithelial type 1 innate lymphoid cells are a unique subset of IL-12- and IL-15-responsive IFN- γ -producing cells. *Immunity.* 2013;38:769–781.
27. Bernink JH, Peters CP, Munneke M, et al. Human type 1 innate lymphoid cells accumulate in inflamed mucosal tissues. *Nat Immunol.* 2013;14:221–229.
28. Price AE, Liang HE, Sullivan BM, et al. Systemically dispersed innate IL-13- expressing cells in type 2 immunity. *Proc Natl Acad Sci USA.* 2010;107:11489–11494.
29. Mjosberg JM, Trifari S, Crellin NK, et al. Human IL-25- and IL-33responsive type 2 innate lymphoid cells are defined by expression of CRTH2 and CD161. *Nat Immunol.* 2011;12:1055–1062.
30. Buonocore S, Ahern PP, Uhlig HH, et al. Innate lymphoid cells drive interleukin-23-dependent innate intestinal pathology. *Nature.* 2010;464:1371–1375.
31. Cella M, Fuchs A, Vermi W, et al. A human natural killer cell subset provides an innate source of IL-22 for mucosal immunity. *Nature.* 2009;457:722–725.
32. Hazenberg MD, Spits H. Human innate lymphoid cells. *Blood.* 2014;124:700–709.
33. Wolk K, Witte E, Witte K, et al. Biology of interleukin-22. *Semin Immunopathol.* 2010;32:17–31.
34. Hanash AM, Dudakov JA, Hua G, et al. Interleukin-22 protects intestinal stem cells from immune-mediated tissue damage and regulates sensitivity to graft versus host disease. *Immunity.* 2012;37:339–350.
- **Role of IL-22 in GVHD incidence and severity.**
35. Munneke JM, Bjorklund AT, Mjosberg JM, et al. Activated innate lymphoid cells are associated with a reduced susceptibility to graft-versus-host disease. *Blood.* 2014;124:812–821.
36. Spielberger R, Stiff P, Bensinger W, et al. Palifermin for oral mucositis after intensive therapy for hematologic cancers. *N Engl J Med.* 2004;351:2590–2598.
37. Tsirigotis P, Triantafyllou K, Girkas K, et al. Keratinocyte growth factor is effective in the prevention of intestinal mucositis in patients with hematological malignancies treated with high-dose chemotherapy and autologous hematopoietic SCT: a video-capsule endoscopy study. *Bone Marrow Transplant.* 2008;42:337–343.
38. Blazar BR, Weisdorf DJ, Defor T, et al. Phase 1/2 randomized, placebo-control trial of palifermin to prevent graft-versus-host disease (GVHD) after allogeneic hematopoietic stem cell transplantation (HSCT). *Blood.* 2006;108:3216–3222.
39. Lindemans CA, Calafiore M, Mertelsmann AM, et al. Interleukin-22 promotes intestinal stem cell mediated epithelial regeneration. *Nature.* 2015;528:560–564.
40. Schwab L, Goroncy L, Palaniyandi S, et al. Neutrophil granulocytes recruited upon translocation of intestinal bacteria enhance GvHD via tissue damage. *Nat Med.* 2014;20:648–654.
41. van der Velden WJ, Netea MG, de Haan AF, et al. Role of the mycobiome in human acute graft-versus-host disease. *Biol Blood Marrow Transplant.* 2013;19:329–332.
42. Nobuhiko K, Sang-Uk S, Grace Y, et al. Role of the gut microbiota in immunity and inflammatory disease. *Nat Rev Immunol.* 2013;13:321–335.
43. Perera PY, Lichy JH, Waldmann TA, et al. The role of interleukin-15 in inflammation and immune responses to infection: implications for its therapeutic use. *Microbes Infect.* 2012;14:247–261.
44. Ferrara JL, Levine JE, Reddy P, et al. “Graft-versus-host disease”. *Lancet.* 2009;373:1550–1561.
45. Steinman RM. Decisions about dendritic cells: past, present, and future. *Annu Rev Immunol.* 2012;30:1–22.
46. Coombes JL, Powrie F. Dendritic cells in intestinal immune regulation. *Nat Rev Immunol.* 2008;8:435–446.
47. Poltorak A, He X, Smirnova I, et al. Defective LPS signaling in C3H/HeJ and C57BL/10ScCr mice: mutations in Tlr4 gene. *Science.* 1998;282:2085–2088.
48. Cook DN, Pisetsky DS, Schwartz DA. Toll-like receptors in the pathogenesis of human disease. *Nat Immunol.* 2004;5:975–979.
49. Nestel FP, Price KS, Seemayer TA, et al. Macrophage priming and lipopolysaccharide triggered release of tumor necrosis factor alpha during graft-versus-host disease. *J Exp Med.* 1992;175:405–413.
50. Cooke KR, Gerbitz A, Crawford JM, et al. LPS antagonism reduces graft-versus-host disease and preserves graft-versus-leukemia activity after experimental bone marrow transplantation. *J Clin Invest.* 2001;107:1581–1589.
51. Cooke KR, Hill GR, Crawford JM, et al. Tumor necrosis factor- α production to lipopolysaccharide stimulation by donor cells predicts the severity of experimental acute graft-versus-host disease. *J Clin Invest.* 1998;102:1882–1891.
52. Fegan C, Poynton CH, Whittaker JA. The gut mucosal barrier in bone marrow transplantation. *Bone Marrow Transplant.* 1990;5:373–377.
53. Shlomchik WD. Graft-versus-host disease. *Nat Rev.* 2007;7:340–352.
54. Calcaterra C, Sfondrini L, Rossini A, et al. Critical role of TLR9 in acute graft-versus-host disease. *J Immunol.* 2008;181:6132–6139.
55. Lorenz E, Schwartz DA, Martin PJ, et al. Association of TLR4 mutations and the risk for acute GVHD after HLA-matched-sibling hematopoietic stem cell transplantation. *Biol Blood Marrow Transplant.* 2001;7:384–387.
56. Elmaagacli AH, Koldehoff M, Hindahl H, et al. Mutations in innate immune system NOD2/CARD 15 and TLR-4 (Thr399Ile) genes influence the risk for severe acute graft-versus-host disease in patients who underwent an allogeneic transplantation. *Transplantation.* 2006;81:247–254.
57. Elmaagacli AH, Steckel N, Ditschkowski M, et al. Toll-like receptor 9, NOD2 and IL23R gene polymorphisms influenced outcome in AML patients transplanted from HLA-identical sibling donors. *Bone Marrow Transplant.* 2011;46:702–708.
58. Highfill SL, Rodriguez PC, Zhou Q, et al. Bone marrow myeloid-derived suppressor cells (MDSCs) inhibit graft-versus-host disease (GvHD) via an arginase1-dependent mechanism that is up-regulated by interleukin-13. *Blood.* 2010;116:5738–5747.
59. Hossain MS, Jaye DL, Pollack BP, et al. Flagellin, a TLR5 agonist, reduces graft-versus-host disease in allogeneic hematopoietic stem cell transplantation recipients while enhancing antiviral immunity. *J Immunol.* 2011;187:5130–5140.
60. Jaspersen LK, Bucher C, Panoskatsis-Mortari A, et al. Inducing the tryptophan catabolic pathway, indoleamine 2,3-dioxygenase (IDO), for suppression of graft-versus-host disease (GvHD) lethality. *Blood.* 2009;114:5062–5070.
61. Lowe EL, Crother TR, Rabizadeh S, et al. Toll-like receptor 2 signaling protects mice from tumor development in a mouse model of colitis-induced cancer. *Plos One.* 2010;5:e13027.
62. O'Brien K, Fitzgerald D, Rostami A, et al. The TLR7 agonist, imiquimod, increases IFN- β production and reduces the severity of experimental autoimmune encephalomyelitis. *J Neuroimmunol.* 2010;221:107–111.

63. Caruso R, Warner N, Inohara N, et al. NOD1 and NOD2: signaling, host defense, and inflammatory disease. *Immunity*. 2014;41:898–908.
64. Penack O, Smith OM, Cunningham-Bussell A, et al. NOD2 regulates hematopoietic cell function during graft-versus-host disease. *J Exp Med*. 2009;206:2101–2110.
- **Recent study showing the inhibitory effect of NOD2 on GVHD.**
65. Watanabe T, Kitani A, Murray PJ, et al. Nucleotide binding oligomerization domain 2 deficiency leads to dysregulated TLR2 signaling and induction of antigen-specific colitis. *Immunity*. 2006;25:473–485.
66. Strober W, Kitani A, Fuss I, et al. The molecular basis of NOD2 susceptibility mutations in Crohn's disease. *Mucosal Immunol*. 2008;1:S5–9.
67. Holler E, Rogler G, Brenmoehl J, et al. Prognostic significance of NOD2/CARD15 variants in HLA identical sibling hematopoietic stem cell transplantation: effect on long-term outcome is confirmed in 2 independent cohorts and may be modulated by the type of gastrointestinal decontamination. *Blood*. 2006;107:4189–4193.
68. Holler E, Rogler G, Herfarth H, et al. Both donor and recipient NOD2/CARD15 mutations associate with transplant-related mortality and GvHD following allogeneic stem cell transplantation. *Blood*. 2004;104:889–894.
69. Lee YK, Mazmanian SK. Has the microbiota played a critical role in the evolution of the adaptive immune system? *Science*. 2010;330:1768–1773.
70. Marchesi JR, Adams DH, Fava F, et al. The gut microbiota and host health: a new clinical frontier. *Gut*. 2016;65:330–339.
71. Rajilic-Stojanovic M, de Vos WM. The first 1000 cultured species of the human gastrointestinal microbiota. *FEMS Microbiol Rev*. 2014;38:996–1047.
72. Shreiner AB, Kao JY, Young VB. The gut microbiome in health and in disease. *Curr Opin Gastroenterol*. 2015;31:6975.
73. Tomkovich S, Jobin C. Microbiota and host immune responses: a love-hate relationship. *Immunology*. 2016;147:1–10.
74. Ivanov II, Frutos Rde L, Manel N, et al. Specific microbiota direct the differentiation of IL-17-producing T-helper cells in the mucosa of the small intestine. *Cell Host Microbe*. 2008;4:337–349.
75. Ivanov II, Atarashi K, Manel N, et al. Induction of intestinal Th17 cells by segmented filamentous bacteria. *Cell*. 2009;139:485–498.
76. Gaboriau-Routhiau V, Rakotobe S, Lécuyer E, et al. The key role of segmented filamentous bacteria in the coordinated maturation of gut helper T cell responses. *Immunity*. 2009;31:677–689.
77. Atarashi K, Tanoue T, Shima T, et al. Induction of colonic regulatory T cells by indigenous *Clostridium* species. *Science*. 2011;331:337–341.
78. Geuking MB, Cahenzli J, Lawson MA, et al. Intestinal bacterial colonization induces mutualistic regulatory T cell responses. *Immunity*. 2011;34:794–806.
79. Round JL, Lee SM, Li J, et al. The Toll-like receptor 2 pathway establishes colonization by a commensal of the human microbiota. *Science*. 2011;332:974–977.
80. Fagarasan S, Muramatsu M, Suzuki K, et al. Critical roles of activation-induced cytidine deaminase in the homeostasis of gut flora. *Science*. 2002;298:1424–1427.
81. Wei M, Shinkura R, Doi Y, et al. Mice carrying a knock-in mutation of *Aicda* resulting in a defect in somatic hypermutation have impaired gut homeostasis and compromised mucosal defense. *Nat Immunol*. 2011;12:264–270.
82. Fagarasan S, Kawamoto S, Kanagawa O, et al. Adaptive immune regulation in the gut: T cell-dependent and T cell-independent IgA synthesis. *Annu. Rev. Immunol*. 2010;28:243–273.
83. Lawley TD, Clare S, Walker AW, et al. Antibiotic treatment of *Clostridium difficile* carrier mice triggers a supershedder state, spore-mediated transmission, and severe disease in immunocompromised hosts. *Infect. Immun*. 2009;77:3661–3669.
84. Rupnik M, Wilcox MH, Gerding DN. *Clostridium difficile* infection: new developments in epidemiology and pathogenesis. *Nature Rev. Microbiol*. 2009;7:526–536.
85. Momose Y, Hirayama K, Itoh K. Competition for proline between indigenous *Escherichia coli* and *E. coli* O157: h7 in gnotobiotic mice associated with infant intestinal microbiota and its contribution to the colonization resistance against *E. coli* O157: H7. *Antonie Van Leeuwenhoek*. 2008;94:165–171.
86. Fabich AJ, Jones SA, Chowdhury FZ, et al. Comparison of carbon nutrition for pathogenic and commensal *Escherichia coli* strains in the mouse intestine. *Infect. Immun*. 2008;76:1143–1152.
87. Leatham MP, Banerjee S, Autieri SM, et al. Precolonized human commensal *Escherichia coli* strains serve as a barrier to *E. coli* O157: h7 growth in the streptomycin-treated mouse intestine. *Infect. Immun*. 2009;77:2876–2886.
88. Bergstrom KS, Kisoos-Singh V, Gibson DL, et al. Muc2 protects against lethal infectious colitis by disassociating pathogenic and commensal bacteria from the colonic mucosa. *Plos Pathog*. 2010;6:e1000902.
89. Petersson J, Schreiber O, Hansson GC, et al. Importance and regulation of the colonic mucus barrier in a mouse model of colitis. *Am. J. Physiol. Gastrointest. Liver Physiol*. 2010;300:G327–G333.
90. Taur Y, Xavier JB, Lipuma L, et al. Intestinal domination and the risk of bacteremia in patients undergoing allogeneic hematopoietic stem cell transplantation. *Clin Infect Dis*. 2012;55:905–914.
- **Large prospective trial assessing microbial diversity with infection and NRM after allo-HSCT.**
91. Vossen J, Heidt P, Guiot H, et al. Prevention of acute graft versus host disease in clinical bone marrow transplantation: complete versus selective intestinal decontamination. Recent advances in germ free research. Tokyo: TUP; 2013; 573–577.
92. Storb R, Prentice RL, Buckner CD, et al. Graft-versus-host disease and survival in patients with aplastic anemia treated by marrow grafts from HLA-identical siblings Beneficial effect of a protective environment. *N Engl J Med*. 1983;308:302–307.
93. Jones JM, Wilson R, Bealmeier PM. Mortality and gross pathology of secondary disease in germfree mouse radiation chimeras. *Radiat Res*. 1971;45:577–588.
94. van Bekkum DW, Roodenburg J, Heidt PJ, et al. Mitigation of secondary disease of allogeneic mouse radiation chimeras by modification of the intestinal microflora. *J Natl Cancer Inst*. 1974;52:401–404.
95. Storb R, Thomas ED. Graft-versus-host disease in dog and man: the Seattle experience. *Immunol Rev*. 1985;88:215–238.
96. Jenq RR, Ubeda C, Taur Y, et al. Regulation of intestinal inflammation by microbiota following allogeneic bone marrow transplantation. *J Exp Med*. 2012;209:903–911.
- **Recent study demonstrating the association of intestinal GVHD with major shifts of intestinal microbiota composition.**
97. Jenq RR, Taur Y, Devlin SM, et al. Intestinal blautia is associated with reduced death from graft-versus-host disease. *Biol Blood Marrow Transplant*. 2015;21:1373–1383.
98. Ganapathy V, Thangaraju M, Prasad PD, et al. Transporters and receptors for short-chain fatty acids as the molecular link between colonic bacteria and the host. *Curr. Opin. Pharmacol*. 2013;13:869–874.
99. Cook SI, Sellin JH. Review article: short chain fatty acids in health and disease. *Aliment. Pharmacol. Ther*. 1998;12:499–507.
100. Wong JMW, De Souza R, Kendall CWC, et al. Colonic health: fermentation and short chain fatty acids. *J. Clin. Gastroenterol*. 2006;40:235–243.
101. Mathewson ND, Jenq R, Mathew AV, et al. Gut microbiome-derived metabolites modulate intestinal epithelial cell damage and mitigate graft-versus-host disease. *Nat Immunol*. 2016;17:505–513.
- **Recent study describing the mitigating role of butyrate – a microbial metabolite on GVHD.**
102. Vossen JM, Heidt PJ, van den Berg H, et al. Prevention of infection and graft-versus-host disease by suppression of intestinal microflora in children treated with allogeneic bone marrow transplantation. *Eur J Clin Microbiol Infect Dis*. 1990;9:14–23.
103. Vossen JM, Guiot HF, Lankester AC. Complete suppression of the gut microbiome prevents acute graft-versus-host disease following allogeneic bone marrow transplantation. *Plos One*. 2014;9:e105706.
104. Docampo MD, Auletta JJ, Jenq RR. Emerging influence of the intestinal microbiota during allogeneic hematopoietic cell

- transplantation: control the gut and the body will follow. *Biol Blood Marrow Transplant*. 2015;21:1360–1366.
105. Taur Y, Jenq RR, Perales MA, et al. The effects of intestinal tract bacterial diversity on mortality following allogeneic hematopoietic stem cell transplantation. *Blood*. 2014;124:1174–1182.
 106. Lozupone CA, Stombaugh JI, Gordon JI, et al. Diversity, stability and resilience of the human gut microbiota. *Nature*. 2012;489:220–230.
 107. Gerbitz A, Schultz M, Wilke A, et al. Probiotic effects on experimental graft-versus-host disease: let them eat yogurt. *Blood*. 2004;103:4365–4367.
 108. Ladas EJ, Bhatia M, Chen L, et al. The safety and feasibility of probiotics in children and adolescents undergoing hematopoietic cell transplantation. *Bone Marrow Transplant*. 2016;51:262–266.
 109. Schrezenmeir J, de Vrese M. Probiotics, prebiotics, and synbiotics—approaching a definition. *Am J Clin Nutr*. 2001;73:361S–364S.
 110. Newburg DS, He Y. Neonatal gut microbiota and human milk glycans cooperate to attenuate infection and inflammation. *Clin Obstet Gynecol*. 2015;58:814–826.
 111. De Castro CG, Ganc AJ, Ganc RL, et al., Fecal microbiota transplant after hematopoietic SCT: report of a successful case. *Bone marrow transplant* 2015; 50: 145.
 112. Bilinski J, Grzesiowski P, Muszynski J, et al. Fecal microbiota transplantation inhibits multidrug-resistant gut pathogens: preliminary report performed in an immunocompromised host. *Arch Immunol Ther Exp (Warsz)* 2016; 64: 255–258.